

Introduction to Cartan's formalism

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Abstract

These notes were presented at the Summer School of Gravitation and General Relativity, held at Centro de Estudios Científicos (CECs), Valdivia, Chile, during March 9th to 13th 2020. Their aim is to give an introduction to Cartan's formalism for graduate students in early stages of their careers. The manuscript contains basic operations in the language of differential forms and the definition of geometrical objects that will be useful in a modern treatment of theoretical physics. The Einstein–Cartan theory of gravity is presented and it is shown how the spin density of matter acts as source of torsion.

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CONTENTS

I. Introduction	3
II. Differential forms	4
A. Wedge product	4
B. Exterior derivative	5
C. Inner contraction	5
D. Lie derivative	6
III. Vielbein, Lorentz connection, curvature, and torsion	7
A. The vielbein	7
B. Inner contraction along vector basis	8
C. The Lorentz connection	10
D. Volume element	10
E. Hodge dual	11
F. Cartan's structure equations	12
G. Levi-Civita connection and contorsion	12
H. Infinitesimal transformations	13
IV. Gravitation: four-dimensional Einstein–Cartan theory	15
A. Action principle	15
B. Field equations	15
C. Torsion from spin	17
V. Discussion	18
Acknowledgments	19
A. Exercises	20
1. Properties	20
2. Hodge dual	20
3. Bianchi identities	20
4. Einstein–Cartan equation in components	20
5. Final project	20

B. Local translations algebra	21
References	22

I. INTRODUCTION

The exterior calculus of differential forms was developed by Élie Cartan in Refs. [1–4] during the 1920s. It provides a powerful tool for describing gravitational and gauge theories in a coordinate independent fashion. The geometry involved in this formalism is known as Riemann–Cartan and it is characterized by two independent objects: the curvature and torsion. Due to their unfamiliar language for physicists at that time, it was only until forty years later that the pioneers works on gauge theories of gravitation reconsidered it [5–7].

The simplest theory in this framework is known as the Einstein–Cartan(–Sciama–Kibble) theory of gravity, in which torsion is induced by the spin density of matter. The absence of the latter degenerates the field equations into general relativity. In fact, it allows one to incorporate the spin density as a gravitational source in the same footing as the energy density of matter fields. The effect of torsion is rather small in this setup, since it appears as a nonpropagating field proportional to the Newton’s constant. Therefore, one would need a huge amount spin density [7] or highly polarized matter [8] to detect it experimentally. However, the cosmological evolution of spin fluids could lead to the avoidance of initial singularities (for a review see [9]). Nevertheless, theories beyond Einstein–Cartan possessing higher-curvature terms [10–12] or nonminimally coupled scalar fields [13–15] could lead to propagating torsion in vacuum.

These lecture notes are not intended to give an exhausting review of the literature nor to replace the excellent books on the subject. Rather, they are meant to provide a self-contained introduction for graduate students in early stages of their careers. They are based on Refs. [16–21] and the readers are encouraged to examine them.

The organization of the manuscript is as follows: in Sec. II, we provide the basic mathematical tools that will be useful from thereon. In Sec. III, we introduce the concept of vielbein, Lorentz connection, curvature, and torsion, as well as their transformation laws. In Sec. IV, we present the Einstein–Cartan theory of gravity and show how the spin density of matter acts as source of torsion. Finally, in Appendix A we include some excercises.

II. DIFFERENTIAL FORMS

The language of differential forms is an appealing approach to modern theoretical physics. It provides a coordinate-independent description of gravitational and gauge theories, that characterize the four interactions in nature. In this section, we review the basic operations within Cartan's exterior calculus, following closely Refs. [16, 20].

A. Wedge product

A differential p -form is a completely antisymmetric tensor of type $(0, p)$ which lies on $\otimes^p T_x^*(\mathcal{M})$, where $T_x^*(\mathcal{M})$ is the cotangent space of the manifold $\mathcal{M}$ at the point x . Its basis is constructed through the wedge product, $\wedge$, that defines a completely antisymmetric tensor product according to

$$dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p} = \sum_{\sigma} (-1)^{|\sigma|} dx^{\sigma(\mu_1)} \otimes \dots \otimes dx^{\sigma(\mu_p)}, \quad (1)$$

where σ denotes the permutation of indices. For instance,

$$\begin{aligned} dx^\mu \wedge dx^\nu &= dx^\mu \otimes dx^\nu - dx^\nu \otimes dx^\mu, \\ dx^\mu \wedge dx^\nu \wedge dx^\lambda &= dx^\mu \otimes dx^\nu \otimes dx^\lambda - dx^\mu \otimes dx^\lambda \otimes dx^\nu - dx^\nu \otimes dx^\mu \otimes dx^\lambda \\ &\quad + dx^\nu \otimes dx^\lambda \otimes dx^\mu + dx^\lambda \otimes dx^\mu \otimes dx^\nu - dx^\lambda \otimes dx^\nu \otimes dx^\mu, \end{aligned} \quad (2)$$

are the basis of the 2 and 3-forms, respectively, and so on.

The wedge product defined in Eq. (1) defines a basis of the vector space of the differential p -forms, denoted by $\Omega^p(\mathcal{M})$. Therefore, an element $\alpha \in \Omega^p(\mathcal{M})$ can be written in this basis according to

$$\alpha = \frac{1}{p!} \alpha_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}. \quad (3)$$

Let us consider $\alpha \in \Omega^p(\mathcal{M})$, $\beta \in \Omega^q(\mathcal{M})$ and $\gamma \in \Omega^r(\mathcal{M})$. Then, the wedge product satisfies the following properties

$$dx^{\mu_1} \wedge \dots \wedge dx^{\mu_s} = 0 \quad \text{if some index } \mu \text{ is repeated,} \quad (4a)$$

$$\alpha \wedge \alpha = 0 \quad \text{if } p \text{ is odd,} \quad (4b)$$

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha, \quad (4c)$$

$$(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma). \quad (4d)$$

B. Exterior derivative

The exterior derivative, d , is a map $d : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p+1}(\mathcal{M})$ which acts on $\alpha \in \Omega^p(\mathcal{M})$ as

$$d\alpha = \frac{1}{p!} \partial_\rho \alpha_{\mu_1 \dots \mu_p} dx^\rho \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}. \quad (5)$$

For instance, if we write the Maxwell potential as $A = A_\mu dx^\mu$, we can use the exterior derivative to define their field strength 2-form as $F = dA$, since

$$F = dA = \partial_\mu A_\nu dx^\mu \wedge dx^\nu = \frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu, \quad (6)$$

where in the last equality the antisymmetric property of the wedge product has been used, i.e. $dx^\mu \wedge dx^\nu = \frac{1}{2}(dx^\mu \wedge dx^\nu - dx^\nu \wedge dx^\mu)$.

If we consider $\alpha \in \Omega^p(\mathcal{M})$, $\beta \in \Omega^q(\mathcal{M})$, and $\gamma \in \Omega^D(\mathcal{M})$, with $\dim \mathcal{M} = D$, the exterior derivative satisfies the following properties

$$d(d\alpha) = d^2\alpha = 0, \quad (7a)$$

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta, \quad (7b)$$

$$d\gamma = 0. \quad (7c)$$

Equation (7b) stems from the fact that the exterior derivative satisfies a graded Leibniz rule, whose sign depends on the degree of the differential forms involved.

If $\alpha \in \Omega^p(\mathcal{M})$ with $d\alpha = 0$, then α is said to be *closed*. If $\beta \in \Omega^q(\mathcal{M})$ that can be expressed as $\beta = d\chi$, where $\chi \in \Omega^{q-1}(\mathcal{M})$, then β is called *exact*. By virtue of the nilpotency property defined in Eq. (7a), any exact form is closed. The Poincaré lemma implies that, locally, any closed form can be expressed as an exact form, but globally it will depend on the topology.

C. Inner contraction

The inner contraction along a vector field V , denoted by i_V , is a map $i_V : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p-1}(\mathcal{M})$ acting on the basis of differential p -form as

$$\begin{aligned} i_V(dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}) &= V^{\mu_1} dx^{\mu_2} \wedge \dots \wedge dx^{\mu_p} - V^{\mu_2} dx^{\mu_1} \wedge dx^{\mu_3} \wedge \dots \wedge dx^{\mu_p} \\ &\quad + \dots + (-1)^{p-1} V^{\mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_{p-1}}, \end{aligned} \quad (8)$$

where the component of the vector field are defined through $V = V^\mu \partial_\mu$. In the same way, the inner contraction acts on $\alpha \in \Omega^p(\mathcal{M})$ as

$$i_V \alpha = \frac{1}{(p-1)!} V^\rho \alpha_{\rho\mu_2\ldots\mu_p} dx^{\mu_2} \wedge \ldots \wedge dx^{\mu_p}. \quad (9)$$

For $\alpha \in \Omega^p(\mathcal{M})$, $\beta \in \Omega^q(\mathcal{M})$, $\phi \in \Omega^0(\mathcal{M})$, namely $\phi : \mathcal{M} \rightarrow \mathbb{R}$ is an scalar function, and V a vector fields, the inner contraction satisfies

$$i_V i_V \alpha = 0, \quad (10a)$$

$$i_V (\alpha \wedge \beta) = i_V \alpha \wedge \beta + (-1)^p \alpha \wedge i_V \beta, \quad (10b)$$

$$i_V \phi = 0, \quad (10c)$$

similar to the properties of the exterior derivative in the previous section.

D. Lie derivative

The Lie derivative is an important operator in differential geometry, since it measures how a tensor changes under the flow generated by a vector field on the manifold. The Lie derivative is a map $\mathcal{L}_V : \Omega^p(\mathcal{M}) \rightarrow \Omega^p(\mathcal{M})$, that can be expressed by the Cartan's formula

$$\mathcal{L}_V = di_V + i_V d. \quad (11)$$

For $\alpha \in \Omega^p(\mathcal{M})$ and $\beta \in \Omega^q(\mathcal{M})$, the Lie derivative satisfies

$$\mathcal{L}_V (\alpha \wedge \beta) = \mathcal{L}_V \alpha \wedge \beta + \alpha \wedge \mathcal{L}_V \beta. \quad (12)$$

Notice that there is no graded Leibniz rule in this case, namely, it does not produce a sign's change when acting on the exterior product of differential forms. Moreover, the Lie derivative of $A \in \Omega^1(\mathcal{M})$ is

$$\begin{aligned} \mathcal{L}_V A &= di_V A + i_V dA = d(V^\mu A_\mu) + i_V (\partial_\mu A_\nu dx^\mu \wedge dx^\nu), \\ &= (\partial_\nu V^\mu A_\mu + V^\mu \partial_\nu A_\mu) dx^\nu + V^\mu \partial_\mu A_\nu dx^\nu - V^\nu \partial_\mu A_\nu dx^\mu, \\ &= (V^\mu \partial_\mu A_\nu + \partial_\nu V^\mu A_\mu) dx^\nu. \end{aligned} \quad (13)$$

This is the usual Lie derivative acting on 1-forms. Moreover, this formula holds for arbitrary p -forms and its computation is straightforward.

The the exterior derivative, inner contraction, and Lie derivative form a graded Lie algebra that, acting on $\alpha \in \Omega^p(\mathcal{M})$, can be written as [16]

$$[\mathcal{L}_U, \mathcal{L}_V] \alpha = (\mathcal{L}_U \mathcal{L}_V - \mathcal{L}_V \mathcal{L}_U) \alpha = \mathcal{L}_{[U, V]} \alpha, \quad (14)$$

$$\{i_U, i_V\} \alpha = (i_U i_V + i_V i_U) \alpha = 0, \quad (15)$$

$$[d, d] \alpha = \{d, d\} \alpha = 0, \quad (16)$$

$$[\mathcal{L}_U, d] \alpha = (\mathcal{L}_U d - d \mathcal{L}_U) \alpha = 0, \quad (17)$$

$$\{i_U, d\} \alpha = (i_U d + d i_U) \alpha = \mathcal{L}_U \alpha, \quad (18)$$

$$[\mathcal{L}_U, i_V] \alpha = (\mathcal{L}_U i_V - i_V \mathcal{L}_U) \alpha = i_{[U, V]} \alpha, \quad (19)$$

where $[U, V]$ denotes the Lie bracket between U and V , that is

$$[U, V] = (U^\mu \partial_\mu V^\nu - V^\mu \partial_\mu U^\nu) \partial_\nu. \quad (20)$$

III. VIELBEIN, LORENTZ CONNECTION, CURVATURE, AND TORSION

A. The vielbein

The equivalence principle allows us to choose a coordinate system in which the spacetime looks locally as Minkowski. This choice can be implemented through the vielbein $e^a_\mu(x)$ as

$$g_{\mu\nu}(x) = e^a_\mu(x) \eta_{ab} e^b_\nu(x), \quad (21)$$

where $\eta_{ab} = \text{diag}(-, +, \dots, +)$ is the Minkowski metric. From hereon, latin and greek characters denote Lorentz and spacetime indices, respectively.

Looking at the left-hand side of Eq. (21), one observes that there are $D(D+1)/2$ independent components of the metric, since it represents a symmetric rank-2 tensor in D -dimensions. The vielbein, on the other hand, has no symmetries on their indices whatsoever, and it has therefore D^2 independent components. This means that, given a metric, the choice of the vielbein is not unique. Nevertheless, the difference in their independent components is exactly $D^2 - D(D+1)/2 = D(D-1)/2$, which matches precisely the number of generators of the local Lorentz group in D -dimensions. Therefore, all the equivalent choices of the vielbein are related by $\Lambda^a_b(x) \in SO(1, D-1)$ as

$$e'^a_\mu(x) = \Lambda^a_b(x) e^b_\mu(x). \quad (22)$$

Additionally, Eq. (21) requires that the vielbein field transforms as a 1-form under general coordinate transformations, that is

$$e'^a{}_\mu(x') = \frac{\partial x^\nu}{\partial x'^\mu} e^a{}_\nu(x). \quad (23)$$

If the vielbein is a non-singular matrix, i.e. $\det e^a{}_\mu \neq 0$, then there exist an inverse vielbein field $E^\mu{}_a$ such that $E^\mu{}_a e^b{}_\mu = \delta^a_b$ and $E^\mu{}_a e^a{}_\nu = \delta^\mu_\nu$. The inverse vielbein transforms according to

$$E'^\mu{}_a(x) = \Lambda^b{}_a E^\mu{}_b \quad \text{and} \quad E'^\mu{}_a(x') = \frac{\partial x'^\mu}{\partial x^\nu} E^\nu{}_a(x), \quad (24)$$

under local Lorentz transformations and general coordinate transformations, respectively. Using the inverse vielbein, it is possible to rewrite the Eq. (21) in terms of

$$E^\mu{}_a(x) g_{\mu\nu}(x) E^\nu{}_b(x) = \eta_{ab}. \quad (25)$$

In fact, it is possible to change the coordinate basis of a type- (p, q) spacetime tensor to a Lorentz one, and vice versa, via

$$T^{a_1 \dots a_p}{}_{b_1 \dots b_q} = e^{a_1}{}_{\mu_1} \dots e^{a_p}{}_{\mu_p} E^{\nu_1}{}_{b_1} \dots E^{\nu_q}{}_{b_q} T^{\mu_1 \dots \mu_p}{}_{\nu_1 \dots \nu_q}, \quad (26)$$

$$T^{\mu_1 \dots \mu_p}{}_{\nu_1 \dots \nu_q} = E^{\mu_1}{}_{a_1} \dots E^{\mu_p}{}_{a_p} e^{b_1}{}_{\nu_1} \dots e^{b_q}{}_{\nu_q} T^{a_1 \dots a_p}{}_{b_1 \dots b_q}. \quad (27)$$

The components of the inverse vielbein field at some point $p \in \mathcal{M}$ with local coordinates x^μ , can be used to construct the vector basis on the tangent space as $E_a = E^\mu{}_a \partial_\mu$. Analogously, the vielbein 1-form defined as

$$e^a = e^a{}_\mu dx^\mu, \quad (28)$$

can be used to construct a new basis for the vector space $\Omega^p(\mathcal{M})$ of differential forms. Thus, the inverse vielbein span an orthonormal basis for vector fields, in the same way as the vielbein 1-form span an orthonormal basis for differential forms. For instance, it is possible to define a differential p -form using

$$\alpha = \frac{1}{p!} \alpha_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p} = \frac{1}{p!} \alpha_{a_1 \dots a_p} e^{a_1} \wedge \dots \wedge e^{a_p}. \quad (29)$$

B. Inner contraction along vector basis

The inner contraction along the vector basis E_a , denoted by $i_a \equiv i_{E_a}$, can be defined as a map $i_a : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p-1}(\mathcal{M})$ from Lorentz tensors to Lorentz tensors, acting on the

orthonormal basis of p -forms as

$$\begin{aligned} i_a (e^{b_1} \wedge \dots \wedge e^{b_p}) &= \delta_a^{b_1} e^{b_2} \wedge \dots \wedge e^{b_p} - \delta_a^{b_2} e^{b_1} \wedge e^{b_3} \wedge \dots \wedge e^{b_p} \\ &\quad + \dots + (-1)^{p-1} \delta_a^{b_p} e^{b_1} \wedge \dots \wedge e^{b_{p-1}}. \end{aligned} \quad (30)$$

For every $\alpha \in \Omega^p(\mathcal{M})$, this operator satisfies

$$\begin{aligned} e^a \wedge i_a \alpha &= e^a \wedge i_a \left(\frac{1}{p!} \alpha_{b_1 \dots b_p} e^{b_1} \wedge \dots \wedge e^{b_p} \right) \\ &= e^a \wedge \left(\frac{p}{p!} \alpha_{ab_2 \dots b_p} e^{b_2} \wedge \dots \wedge e^{b_p} \right) = p \alpha. \end{aligned} \quad (31)$$

Problem: Consider the equation (see Ref. [17])

$$Q_a + L_{ab} \wedge e^b = 0, \quad (32)$$

where $Q_a \in \Omega^2(\mathcal{M})$ and $L_{ab} \in \Omega^1(\mathcal{M})$, with $L_{ab} = -L_{ba}$, whose components are defined via $Q_a = \frac{1}{2} Q_{abc} e^b \wedge e^c$ and $L_{ab} = L_{abc} e^c$. Solve this equation for L_{ab} .

Solution: Applying the inner contraction along the vector basis E_c on $L_{ab} \wedge e^b$ and changing $a \leftrightarrow c$, one has

$$i_c (L_{ab} \wedge e^b) = L_{abc} e^b - L_{ac}, \quad (33)$$

$$i_a (L_{cb} \wedge e^b) = L_{cba} e^b - L_{ca}, \quad (34)$$

respectively. Subtracting these two equations leads to

$$2L_{ac} + (L_{cba} - L_{abc}) e^b = 2i_{[a} (L_{c]b} \wedge e^b), \quad (35)$$

where antisymmetrization of indices includes a normalization factor equal to the number of indices between square brackets, e.g. $A_{[ab]} = \frac{1}{2}(A_{ab} - A_{ba})$. Applying the inner contraction once more on Eq. (33), leads to

$$i_d [i_c (L_{ab} \wedge e^b)] = L_{adc} - L_{acd} = L_{cad} - L_{dac}, \quad (36)$$

where in the last equality we have used $L_{abc} = -L_{bac}$. Therefore, using this result, the second term on the left-hand side of Eq. (35) can be expressed as

$$(L_{cba} - L_{abc}) e^b = [i_a i_c (L_{bd} \wedge e^d)] e^b. \quad (37)$$

Thus, replacing this expression into Eq. (35), one obtains

$$L_{ab} = i_{[a} (L_{b]c} \wedge e^c) - \frac{1}{2} [i_a i_b (L_{cd} \wedge e^d)] e^c. \quad (38)$$

Finally, using $L_{ab} \wedge e^b = -Q_a$ from Eq. (32), we arrive to

$$L_{ab} = -i_{[a} Q_{b]} + \frac{1}{2} (i_a i_b Q_c) e^c. \quad (39)$$

This result will be used in the forthcoming sections.

C. The Lorentz connection

In order to define covariant derivatives under local Lorentz transformations, we need to introduce a gauge connection for such a group. This is called the Lorentz connection 1-form, denoted by $\omega^a_b(x) = \omega^a_{b\mu}(x)dx^\mu$, whose transformation law under $SO(1, D-1)$ is given by

$$\omega^a_b \rightarrow \omega'^a_b = \Lambda^a_c \omega^c_d \Lambda^d_b + \Lambda^a_b d\Lambda^c_d. \quad (40)$$

In this way, we can define the exterior covariant derivative with respect to ω^a_b , denoted by D , as a map $D : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p+1}(\mathcal{M})$ acting on Lorentz tensors $T^{a_1 \dots a_p}_{b_1 \dots b_q} \in \Omega^p(\mathcal{M})$ as

$$\begin{aligned} DT^{a_1 \dots a_p}_{b_1 \dots b_q} &= dT^{a_1 \dots a_p}_{b_1 \dots b_q} + \omega^{a_1}_{c_1} \wedge T^{c_1 \dots a_p}_{b_1 \dots b_q} + \dots + \omega^{a_p}_{c_p} \wedge T^{a_1 \dots c_p}_{b_1 \dots b_q} \\ &\quad - \omega^{c_1}_{b_1} \wedge T^{a_1 \dots a_p}_{c_1 \dots b_q} - \dots - \omega^{c_q}_{b_q} \wedge T^{a_1 \dots a_p}_{b_1 \dots c_q}. \end{aligned} \quad (41)$$

This object transform covariantly under local Lorentz transformations, provided that the Lorentz connection transform as (40).

D. Volume element

The Levi-Civita symbol $\epsilon_{a_1 \dots a_D}$ is an invariant tensor under $SO(1, D-1)$,¹ where

$$\epsilon_{a_1 \dots a_D} = \begin{cases} +1, & \text{for even permutation of } a_1 \dots a_D, \\ -1, & \text{for odd permutation of } a_1 \dots a_D, \\ 0, & \text{if any of the indices apperars repeated at least once.} \end{cases} \quad (42)$$

¹ Henceforth, we use the normalization $\epsilon_{01 \dots D-1} = 1 = -\epsilon^{01 \dots D-1}$.

The volume element D -form, denoted by ϵ , can be defined by means of the Levi-Civita symbol according to

$$\begin{aligned}\epsilon &= \frac{1}{D!} \epsilon_{a_1 \dots a_D} e^{a_1} \wedge \dots \wedge e^{a_D} \\ &= \frac{1}{D!} \epsilon_{a_1 \dots a_D} e^{a_1}_{\mu_1} \dots e^{a_D}_{\mu_D} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D} = e d^D x,\end{aligned}\tag{43}$$

where $e = \det e^a_{\mu}$, $d^D x = \frac{1}{D!} \epsilon_{\mu_1 \dots \mu_D} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_D}$, and we have used the formula of the determinant of a $D \times D$ matrix M^a_b

$$\det M \epsilon_{a_1 \dots a_D} = \epsilon_{m_1 \dots m_D} M^{m_1}_{a_1} \dots M^{m_D}_{a_D}.\tag{44}$$

Moreover, the volume element satisfies $i_{a_D} \dots i_{a_1} \epsilon = \epsilon_{a_1 \dots a_D}$. Using the identity

$$\epsilon^{a_1 \dots a_p c_{p+1} \dots c_D} \epsilon_{b_1 \dots b_p c_{p+1} \dots c_D} = -p!(D-p)! \delta^{a_1}_{[b_1} \dots \delta^{a_p]}_{b_p]},\tag{45}$$

for Lorentzian manifolds, we arrive to

$$e^{a_1} \wedge \dots \wedge e^{a_D} = -\epsilon^{a_1 \dots a_D} e d^D x,\tag{46}$$

which can be regarded as the covariant volume element upon the identification $\sqrt{-g} = e$ from Eq. (21).

E. Hodge dual

If the manifold $\mathcal{M}$ is endowed with a metric $g_{\mu\nu}$ whose inverse exists, the Hodge dual can be defined as a map $\star : \Omega^p(\mathcal{M}) \rightarrow \Omega^{D-p}(\mathcal{M})$, where $\dim \mathcal{M} = D$, acting on the differential form basis as

$$\star(dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}) = \frac{\sqrt{-g}}{(D-p)!} \epsilon^{\mu_1 \dots \mu_p}_{\nu_{p+1} \dots \nu_D} dx^{\nu_{p+1}} \wedge \dots \wedge dx^{\nu_D},\tag{47}$$

$$\text{or} \quad \star(e^{a_1} \wedge \dots \wedge e^{a_p}) = \frac{1}{(D-p)!} \epsilon^{a_1 \dots a_p}_{a_{p+1} \dots a_D} e^{a_{p+1}} \wedge \dots \wedge e^{a_D}.\tag{48}$$

It is worth noticing that, even though $dd\alpha = 0$ for $\alpha \in \Omega^p(\mathcal{M})$, $d\star d\alpha \neq 0$ in general. This implies that introducing the Hodge dual could lead to higher-order differential equations.

F. Cartan's structure equations

The curvature and torsion 2-forms are defined through the Cartan's structure equations, which are given by

$$R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b = \frac{1}{2} \mathcal{R}^{ab}{}_{cd} e^c \wedge e^d, \quad (49)$$

$$T^a = de^a + \omega^a{}_b \wedge e^b = \frac{1}{2} \mathcal{T}^a{}_{bc} e^b \wedge e^c. \quad (50)$$

Moreover, they satisfy the Bianchi identities

$$DR^a{}_b = 0 \quad \text{and} \quad DT^a = R^a{}_b \wedge e^b. \quad (51)$$

The curvature and torsion 2-forms will play an essential role in gravitational theories written in Cartan's language. This stems from the fact that these objects transform as tensors under $SO(1, D-1)$ and they can be used to construct invariant action principles. Even more, they encode the dynamics of the fundamental fields e^a and $\omega^a{}_b$, since they are constructed out of their derivatives.

G. Levi-Civita connection and contorsion

As it is well known, the difference between two connections is a tensor. The Levi-Civita connection, denoted by $\tilde{\omega}^a{}_b$, is defined as the one having vanishing torsion, namely

$$de^a + \tilde{\omega}^a{}_b \wedge e^b = 0. \quad (52)$$

Thus, the contorsion 1-form, denoted by $K^a{}_b$, can be regarded as the difference between the Lorentz connection and the Levi-Civita connection, i.e.

$$K^a{}_b = \omega^a{}_b - \tilde{\omega}^a{}_b. \quad (53)$$

The contorsion 1-form satisfies $K^{ab} = -K^{ba}$, where the latin indices are raised and lowered by the Minkowski metric.

Replacing Eq. (53) into Eqs. (49) and (50), respectively yields to

$$R^a{}_b = \tilde{R}^a{}_b + \tilde{D}K^a{}_b + K^a{}_c \wedge K^c{}_b, \quad (54)$$

$$T^a = K^a{}_b \wedge e^b, \quad (55)$$

where Eq. (52) has been used. Moreover, the quantities $\tilde{R}^a_b = d\tilde{\omega}^a_b + \tilde{\omega}^a_c \wedge \tilde{\omega}^c_b$, alongside $\tilde{D}K^a_b = dK^a_b + \tilde{\omega}^a_c \wedge K^c_b - \tilde{\omega}^c_b \wedge K^a_c$ have been defined. Using Eq. (39), the Levi-Civita connection and the contorsion 1-forms can be solved in terms of the exterior derivative of the vielbein and the torsion, respectively, as

$$\tilde{\omega}_{ab} = -i_{[a}de_{b]} + \frac{1}{2}(i_a i_b de_c)e^c, \quad (56)$$

$$K_{ab} = i_{[a}T_{b]} - \frac{1}{2}(i_a i_b T_c)e^c. \quad (57)$$

H. Infinitesimal transformations

The infinitesimal local Lorentz transformations (LLT) can be obtained by expanding the elements of $SO(1, D-1)$ near the identity as

$$\Lambda^a_b = \delta^a_b + \lambda^a_b + \dots \quad (58)$$

where $|\lambda^a_b| \ll 1$. Replacing this expansion into the definition of the Minkowski metric, i.e. $\Lambda^a_b \eta_{ac} \Lambda^c_d = \eta_{bd}$, and keeping only terms linear in λ^a_b , one concludes that $\lambda_{ab} = -\lambda_{ba}$, with $\lambda_{ab} = \eta_{ac} \lambda^c_b$. The vielbein and Lorentz connection transform infinitesimally under $SO(1, D-1)$ as

$$\text{LLT} : \begin{cases} \delta_{\text{LLT}}(\lambda)e^a &= \lambda^a_b e^b, \\ \delta_{\text{LLT}}(\lambda)\omega^a_b &= -D\lambda^a_b. \end{cases} \quad (59)$$

On the other hand, the Lie derivative is a map between Lorentz-scalar p -forms into Lorentz-scalar p -forms. The presence of the exterior derivative in their definition (11) spoils the Lorentz-covariant character of this operation when applied to Lorentz tensors. However, one can observe that the Lie derivative can be written as a linear combination of LLT and an additional transformation usually called local translations (LT), which are in fact Lorentz covariant.² To see this, let us apply the Lie derivative on e^a and ω^a_b by using Eq. (11), i.e.

$$\begin{aligned} \mathcal{L}_\xi e^a &= di_\xi e^a + i_\xi de^a, \\ &= d\xi^a + i_\xi (T^a - \omega^a_b \wedge e^b), \\ &= d\xi^a + i_\xi T^a - i_\xi \omega^a_b e^b + \omega^a_b \xi^b, \\ &= D\xi^a + i_\xi T^a + \tilde{\lambda}^a_b e^b. \end{aligned}$$

² They are also called Lie draggings or improved diffeomorphisms in the literature.

In the second line, we have used the identity $i_\xi e^a = \xi^a$ and the definition of torsion [c.f. Eq. (50)], in the third line we have used the graded Leibniz rule of the operator i_ξ , and in the last one we have define the 0-form $\tilde{\lambda}^a_b = -i_\xi \omega^a_b$ that mimicks the role of an infinitesimal Lorentz transformation of the vielbein. For the Lorentz connection, on the other hand, we have

$$\begin{aligned}
\mathcal{L}_\xi \omega^a_b &= di_\xi \omega^a_b + i_\xi d\omega^a_b, \\
&= di_\xi \omega^a_b + i_\xi (R^a_b - \omega^a_c \wedge \omega^c_b), \\
&= di_\xi \omega^a_b + i_\xi R^a_b - i_\xi \omega^a_c \omega^c_b + \omega^a_c i_\xi \omega^c_b, \\
&= di_\xi \omega^a_b + \omega^a_c i_\xi \omega^c_b - \omega^c_b i_\xi \omega^a_c + i_\xi R^{ab}, \\
&= i_\xi R^{ab} - D\tilde{\lambda}^{ab},
\end{aligned}$$

where in the second line we have used the definition of the curvature [c.f. Eq. (49)]. Thus, the Lie derivative acting on either the vielbein or the Lorentz connection, can interpreted as a linear combination of local translations plus a local Lorentz transformation given by the parameter $\tilde{\lambda}^a_b = -i_\xi \omega^a_b$, that can be summarized as

$$\mathcal{L}_\xi \begin{pmatrix} e^a \\ \omega^a_b \end{pmatrix} = \delta_{\text{LT}}(\xi) \begin{pmatrix} e^a \\ \omega^a_b \end{pmatrix} + \delta_{\text{LLT}}(\tilde{\lambda}) \begin{pmatrix} e^a \\ \omega^a_b \end{pmatrix}, \quad (60)$$

where the local translations are defined by

$$\text{LT} : \begin{cases} \delta_{\text{LT}}(\xi) e^a = D\xi^a + i_\xi T^a, \\ \delta_{\text{LT}}(\xi) \omega^{ab} = i_\xi R^{ab}. \end{cases} \quad (61)$$

Remarkably, LLT and LT acting on either e^a or ω^{ab} , form an algebroid of local transformations given by [5–7, 22, 23]

$$[\delta_{\text{LLT}}(\lambda_1), \delta_{\text{LLT}}(\lambda_2)] = \delta_{\text{LLT}}(\lambda_3), \quad (62a)$$

$$[\delta_{\text{LLT}}(\lambda), \delta_{\text{LT}}(\xi)] = \delta_{\text{LT}}(\tilde{\xi}), \quad (62b)$$

$$[\delta_{\text{LT}}(\xi_1), \delta_{\text{LT}}(\xi_2)] = \delta_{\text{LLT}}(\bar{\lambda}) + \delta_{\text{LT}}(\bar{\xi}), \quad (62c)$$

where $\lambda_3^{ab} = 2\lambda_1^{[a} \lambda_2^{b]}$, $\tilde{\xi}^a = \lambda^a_b \xi^b$, $\bar{\lambda}^{ab} = i_{\xi_2} i_{\xi_1} R^{ab}$, and $\bar{\xi}^a = i_{\xi_1} i_{\xi_2} T^a$. The computation of the first two commutators is straightforward. The last one, however, is more involved and some identities are given in Appendix B for the interested reader. Since the transformations in Eq. (62c) depends on R^a_b and T^a , the algebra is endowed with structure functions rather than structure constants.

IV. GRAVITATION: FOUR-DIMENSIONAL EINSTEIN–CARTAN THEORY

In this section, we describe the dynamics of Einstein–Cartan theory of gravitation in four dimensions by implementing the Cartan’s formalism described before. To this end, we treat the vielbein and Lorentz connection as independent variables and show how the spin density of matter fields acts as a source of torsion.

A. Action principle

The four-dimensional Einstein–Cartan theory can be described by the action principle

$$I[e^a, \omega^{ab}] = \frac{1}{4\kappa} \int \epsilon_{abcd} \left(R^{ab} - \frac{\Lambda}{6} e^a \wedge e^b \right) \wedge e^c \wedge e^d + \int \mathcal{L}_m[e^a, \omega^{ab}, \Psi], \quad (63)$$

where $\kappa = 8\pi G_N$, Λ is the cosmological constant, $\mathcal{L}_m \in \Omega^4(\mathcal{M})$, and Ψ collectively denote matter fields. The first term on the right-hand side is the usual Einstein–Hilbert action written in differential forms for a torsionful Lorentz connection ω^a_b , since

$$\begin{aligned} \frac{1}{4\kappa} \int \epsilon_{abcd} \left(R^{ab} - \frac{\Lambda}{6} e^a \wedge e^b \right) \wedge e^c \wedge e^d &= \frac{1}{4\kappa} \int \epsilon_{abcd} \left(\frac{1}{2} \mathcal{R}^{ab}{}_{mn} - \frac{\Lambda}{6} \delta_{[m}^a \delta_{n]}^b \right) e^m \wedge e^n \wedge e^c \wedge e^d \\ &= \frac{1}{4\kappa} \int \epsilon_{abcd} \left(\frac{1}{2} \mathcal{R}^{ab}{}_{mn} - \frac{\Lambda}{6} \delta_{[m}^a \delta_{n]}^b \right) (-\epsilon^{mncd} e^4 dx) \\ &= \frac{1}{4\kappa} \int d^4x e (2\delta_{[a}^m \delta_{b]}^n \mathcal{R}^{ab}{}_{mn} - 4\Lambda) \\ &= \frac{1}{2\kappa} \int d^4x e (\mathcal{R} - 2\Lambda), \end{aligned}$$

where in the second line we have used Eq. (46) and in the third one the identity (45). The object $\mathcal{R} = \mathcal{R}^{ab}{}_{ab}$ is the Ricci scalar constructed out of the components of the Lorentz curvature.

B. Field equations

The field equations are obtained by performing stationary variations of the action (63) with respect to the vielbein e^a , Lorentz connection ω^a_b , and matter fields Ψ , treated as independent variables. First, the variation of the action (63) with respect to e^a , denoted by

$\delta_e I$, gives

$$\begin{aligned}\delta_e I &= \frac{1}{4\kappa} \int \epsilon_{abcd} \left(2R^{ab} - \frac{4\Lambda}{6} e^a \wedge e^b \right) \wedge \delta e^c \wedge e^d + \int \underbrace{\frac{\partial \mathcal{L}_m}{\partial e^a}}_{\equiv \tau_a} \wedge \delta e^a \\ &= \frac{1}{2\kappa} \int \delta e^a \left[\epsilon_{abcd} \left(R^{bc} - \frac{\Lambda}{3} e^b \wedge e^c \right) \wedge e^d - 2\kappa \tau_a \right],\end{aligned}\quad (64)$$

where $\frac{\partial \mathcal{L}_m}{\partial e^a}$ denotes the functional derivative of $\mathcal{L}_m$ with respect to the vielbein. Variation of the action (63) with respect to ω^{ab} , denoted by $\delta_\omega I$, gives

$$\delta_\omega I = \frac{1}{2\kappa} \int \epsilon_{abcd} \delta_\omega R^{ab} \wedge e^c \wedge e^d + \int \underbrace{\frac{\partial \mathcal{L}_m}{\partial \omega^{ab}}}_{\equiv \frac{1}{2} \sigma_{ab}} \wedge \omega^{ab} \quad (65)$$

From Eq. (49), it is clear that $\delta_\omega R^{ab} = D\delta\omega^{ab}$. Replacing this expression into $\delta_\omega I$ and integrating by part through

$$\begin{aligned}d(\epsilon_{abcd} \delta\omega^{ab} \wedge e^c \wedge e^d) &= D(\epsilon_{abcd} \delta\omega^{ab} \wedge e^c \wedge e^d) \\ &= \epsilon_{abcd} D\delta\omega^{ab} \wedge e^c \wedge e^d - 2\epsilon_{abcd} \delta\omega^{ab} \wedge T^c \wedge e^d,\end{aligned}\quad (66)$$

we obtain

$$\delta_\omega I = \frac{1}{2\kappa} \int \delta\omega^{ab} [\epsilon_{abcd} T^c \wedge e^d - \kappa \sigma_{ab}] + \text{boundary terms.} \quad (67)$$

Thus, the field equation for the vielbein, Lorentz connection, and matter fields in the four-dimensional Einstein–Cartan theory are

$$\epsilon_{abcd} \left(R^{bc} - \frac{\Lambda}{3} e^b \wedge e^c \right) \wedge e^d = 2\kappa \tau_a, \quad (68)$$

$$\epsilon_{abcd} T^c \wedge e^d = \kappa \sigma_{ab}, \quad (69)$$

$$\frac{\delta \mathcal{L}_m}{\delta \Psi} = 0, \quad (70)$$

respectively. The first equation shows how the energy-momentum 3-form τ_a is source of curvature, in the same way as the second one shows how the spin density 3-form σ_{ab} is source of torsion. The last line is merely the equation of motion for matter fields.

The contracted Bianchi identities of the Einstein–Cartan theory can be obtained by defining the Lorentz-valued 3-forms

$$\mathcal{E}_a = \epsilon_{abcd} \left(R^{bc} - \frac{\Lambda}{3} e^b \wedge e^c \right) \wedge e^d, \quad (71)$$

$$\mathcal{E}_{ab} = \epsilon_{abcd} T^c \wedge e^d. \quad (72)$$

Then, taking the exterior covariant derivative on these quantities and after some algebra, one gets

$$D\mathcal{E}_a = i_a T^b \wedge \mathcal{E}_b + i_a R^{bc} \wedge \mathcal{E}_{bc}, \quad (73)$$

$$D\mathcal{E}_{ab} = e_{[a} \wedge \mathcal{E}_{b]}. \quad (74)$$

Using the equations of motion (68) and (69), the conservation laws of the energy-momentum and spin density 3-forms are given by

$$D\tau_a = i_a T^b \wedge \tau_b + i_a R^{bc} \wedge \sigma_{bc}, \quad (75)$$

$$D\sigma_{ab} = 2e_{[a} \wedge \tau_{b]}, \quad (76)$$

respectively. These identities are consequence of the invariance of the field equations of Einstein–Cartan theory under local translations and local Lorentz transformations.

C. Torsion from spin

Equation (69) is algebraic for the torsion, and it can be solved by contracting both sides with the Levi-Civita symbol ϵ^{ablm} and, using Eq. (45) and renaming dummy indices, gives

$$T^{[a} \wedge e^{b]} = -\frac{\kappa}{4} \epsilon^{abcd} \sigma_{cd}. \quad (77)$$

Then, applying the inner contraction i_b on both sides, one gets

$$\begin{aligned} i_b T^{[a} \wedge e^{b]} + T^{[a} \delta_b^{b]} &= -\frac{\kappa}{4} \epsilon^{abcd} i_b \sigma_{cd} \\ \frac{1}{2} \left(\underbrace{i_b T^a \wedge e^b}_{=-2T^a} - i_b T^b \wedge e^a \right) + \frac{1}{2} (4T^a - T^a) &= -\frac{\kappa}{4} \epsilon^{abcd} i_b \sigma_{cd} \\ \frac{1}{2} T^a - \frac{1}{2} i_b T^b \wedge e^a &= -\frac{\kappa}{4} \epsilon^{abcd} i_b \sigma_{cd}, \end{aligned} \quad (78)$$

where in the second line we have used the property of the inner contraction $e^a \wedge i_a \alpha = p\alpha$, with $\alpha \in \Omega^p(\mathcal{M})$. Contracting both sides of the last equation with i_a once more, gives

$$\begin{aligned} \frac{1}{2} i_a T^a - \underbrace{\frac{1}{2} i_a i_b T^b e^a}_{\frac{1}{2} i_b T^b} + \frac{1}{2} i_b T^b \delta_a^a &= -\frac{\kappa}{4} \epsilon^{abcd} i_a i_b \sigma_{cd} \\ i_b T^b &= -\frac{\kappa}{8} \epsilon^{abcd} i_a i_b \sigma_{cd}. \end{aligned} \quad (79)$$

Replacing this into Eq. (78), one obtains the torsion as a function of the spin density as

$$T^a = -\frac{\kappa}{8}\epsilon^{lmpq}(i_l i_m \sigma_{pq} \wedge e^a + 4\delta_l^a i_m \sigma_{pq}). \quad (80)$$

Thus, this expression and Eq. (57) allows one to determine completely the contorsion 1-form in terms of the spin density. Moreover, the decomposition of the Lorentz connection according to Eq. (53), allows one to rewrite the equation of motion for the vielbein (68) as

$$\epsilon_{abcd} \left(\tilde{R}^{bc} - \frac{\Lambda}{3} e^b \wedge e^c \right) \wedge e^d = 2\kappa \tau_a^{(\text{eff})}, \quad (81)$$

where the effective energy-momentum 3-form $\tau_a^{(\text{eff})}$ contains the information of the spin density through K^{ab} as

$$\tau_a^{(\text{eff})} = \tau_a - \frac{1}{2\kappa} \epsilon_{abcd} \left(\tilde{D} K^{bc} + K^b{}_m \wedge K^{mc} \right) \wedge e^d. \quad (82)$$

Equation (81) represents the Einstein's field equation for an arbitrary vielbein basis, as it can be checked by taking their Hodge dual and identifying $\star \tau_a^{(\text{eff})} = -\tau_{ba}^{(\text{eff})} e^b$. As it is mentioned in Ref. [18], the role of the effective energy-momentum 3-form is similar to the improved Belinfante–Rosenfeld tensor, that it is symmetric and conserved. The contribution of the spin density on the right-hand side of Eq. (82) is relevant for singularity theorems, since it can generate a violation of energy conditions and produce avoidance of singularities [24, 25].

V. DISCUSSION

The Cartan's formalism is an interesting approach to deal with gravity as a gauge theory. In this framework, the gravitational interaction is interpreted as the manifestation of the Riemann–Cartan geometry through curvature and torsion. In their simplest formulation—the Einstein–Cartan theory—, torsion is a nonpropagating field that is sourced by the spin density of matter, whose absence yields to general relativity on shell. Since most of the experimental tests of general relativity are performed in vacuum, this theory is compatible with them and their verification needs for a huge and highly polarized spin density of matter. One would expect that these conditions occur in the interior of neutron stars, providing, in principle, a suitable laboratory for testing torsion.

In theories beyond Einstein–Cartan, propagating torsion induces a change in the polarization of gravitational waves without affecting their speed [26–28]. Thus, future multi-

messenger observations will be crucial for determining the propagating nature of torsion by measuring the change of polarization of gravitational waves.

When Dirac spinors are included, the chiral anomaly introduces the Nieh–Yan invariant [29], that becomes divergent once the regulator needed for the computation is removed [30]. An interesting approach to regularize this divergence is to include a pseudoscalar field nonminimally coupled to the Nieh–Yan density, whose shift symmetry renders the theory finite [31]. This modification has appealing consequences, such as offering a dark matter candidate [32] and providing a gravitational mechanism to solve the strong CP problem [33]. Moreover, the nonminimal coupling of the (pseudo)scalar field to topological invariants supports the existence of four-dimensional homogeneous black string solutions with nontrivial torsion [34] and interesting cosmological solutions [13, 14, 35].

In higher-dimensions, the most general action principle that: (i) is invariant under diffeomorphisms and local Lorentz transformations, (ii) it is constructed out of the vielbein, Lorentz connection and their derivatives, and (iii) it leads to first-order field equations, is the Lovelock–Cartan theory [11]. Here, the presence of higher-curvature terms produce propagating torsion in vacuum. In fact, the maximum number of degrees of freedoms in this theory occur when the coefficients are chosen in such a way that the action can be written as the Chern–Simons form for the (A)dS group or as the gravitational Born–Infeld theory, in odd and even dimensions, respectively [36]. Compatibility with Bianchi identities impose severe restrictions on the torsion components if the coefficients are chosen otherwise. Some exact solutions are known in this framework [37–43].

Finally, it is worth mentioning that relaxing our assumptions on the geometry of space-time could reveal new gravitational phenomena that can be tested in future experiments.

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Appendix A: Exercises

1. Properties

In Sec. II, we give different properties that the basic operations of Cartan's formalism satisfy. Show Eqs. (4), (7), and (10).

2. Hodge dual

Consider a D -dimensional spacetime whose metric has Lorentzian signature $(-, +, \dots, +)$. Show that the Hodge dual, acting on $\alpha \in \Omega^p(\mathcal{M})$, satisfies

$$\star \star \alpha = (-1)^{1+p(D-p)} \alpha. \quad (\text{A1})$$

3. Bianchi identities

Show the Bianchi identities $\text{D}R^{ab} = 0$ and $\text{D}T^a = R^a_b \wedge e^b$. Then, using the definitions of $\mathcal{E}_a$ and $\mathcal{E}_{ab}$ in Eq. (71), show the contracted Bianchi identity (73) by calculating $\text{D}\mathcal{E}_a$, introducing $\delta_a^m = i_a e^m$, and using the graded Leibniz rule of i_a to “integrate by parts”. Then, show Eq. (74) by calculating $\text{D}\mathcal{E}_{ab}$ and introducing $\delta_a^b = -\frac{1}{3}\epsilon_{almn}\epsilon^{blmn}$.

4. Einstein–Cartan equation in components

Consider the field equations (68) and (69) and, by taking the Hodge dual, show that, for an arbitrary vielbein 1-form, one gets

$$\mathcal{R}^a_b - \frac{1}{2}\delta_b^a \mathcal{R} + \Lambda \delta_b^a = \kappa \tau^a_b, \quad (\text{A2})$$

$$\mathcal{T}^a_{bc} + 2\delta^a_{[b} \mathcal{T}_{c]} = \kappa \sigma^a_{bc}, \quad (\text{A3})$$

where $\mathcal{R}^a_b = \mathcal{R}^{am}_{bm}$, $\mathcal{R} = \mathcal{R}^a_a$, $\mathcal{T}_a = \mathcal{T}^b_{ab}$, $\star \tau_a = -\eta_{bc} \tau^b_a e^c$, and $\star \sigma_{ab} = \eta_{cd} \sigma^c_{ab} e^d$.

5. Final project

Consider the four-dimensional Dirac action

$$I_D [e^a, \omega^{ab}, \psi] = \frac{1}{2} \int \left(\bar{\psi} \star \gamma \wedge \text{D}\psi + \text{D}\bar{\psi} \wedge \star \gamma \psi \right), \quad (\text{A4})$$

with $\bar{\psi} = i\psi^\dagger\gamma^0$, $\gamma = \gamma_a e^a$ being the 1-form gamma matrix such that $\{\gamma_a, \gamma_b\} = 2\eta_{ab}$, $D\psi = d\psi + \frac{1}{4}\omega^{ab}\gamma_{ab}\psi$, where $\gamma_{a_1\dots a_p} \equiv \gamma_{[a_1}\dots\gamma_{a_p]}$. Then,

1. Using the fact that $\bar{\psi}\psi$ is a scalar, show that $D\bar{\psi} = d\bar{\psi} - \frac{1}{4}\omega^{ab}\bar{\psi}\gamma_{ab}$.
2. Perform stationary variation of Eq. (A4) with respect to ψ to obtain the Dirac equation in presence of torsion, given by

$$\star\gamma \wedge \left(D\psi - \frac{1}{2}i_a T^a \psi \right) = 0. \quad (\text{A5})$$

3. Show $\delta_e \star\gamma = \delta e^a \wedge \star(\gamma \wedge e_a)$ and $\{\gamma_a, \gamma_{bc}\} = 2i\epsilon_{abcd}\gamma^d\gamma_5$, where $\gamma_5 = i\gamma_0\gamma_1\gamma_2\gamma_3$.
4. From the definition of the energy-momentum and spin density 3-forms, obtain

$$\begin{aligned} \tau_a^{(\psi)} &= \frac{1}{2} \left(D\bar{\psi} \wedge \star(\gamma \wedge e_a) \psi - \bar{\psi} \star(\gamma \wedge e_a) \wedge D\psi \right), \\ \sigma_{ab}^{(\psi)} &= \frac{1}{2} J_5 \wedge e_a \wedge e_b, \end{aligned} \quad (\text{A6})$$

where the 1-form fermionic axial current is defined by $J_5 \equiv i\bar{\psi}\gamma_5\gamma_a\psi e^a$.

5. Compute the torsion 2-form and contorsion 1-form K^{ab} to obtain

$$T^a = \frac{\kappa}{2} \star (J_5 \wedge e^a), \quad (\text{A7})$$

$$K^{ab} = -\frac{1}{4}\kappa \star (J_5 \wedge e^a \wedge e^b). \quad (\text{A8})$$

6. Use Eq. (82) to obtain the contribution of the spin of Dirac spinors to the energy-momentum 3-form.

Appendix B: Local translations algebra

Some identities needed to compute the algebra of LT are obtained in this Appendix. For simplicity, the commutator is given as acting on the vielbein, even though the same identities are useful in the case where the commutator acts on the Lorentz connection. The relation between the Lie derivative, LT, and LLT, given in Eq. (60) implies

$$[\delta_{\text{LT}}(\xi_1), \delta_{\text{LT}}(\xi_2)] e^a = (i_{\xi_1} R^a_b) \xi_2^b + \mathcal{L}_{\xi_1} i_{\xi_2} T^a - (i_{\xi_1} D\xi_2^b) i_b T^a + (i_{\xi_1} \omega^a_b) i_{\xi_2} T^b - (1 \leftrightarrow 2).$$

Now, from the equivalent torsion definition (see Ref. [20]), $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$, where ∇ is a torsion-free covariant derivative and $\mathcal{L}_X Y = [X, Y]$, it follows that

$$i_{\xi_2} i_{\xi_1} T^a + \mathcal{L}_{\xi_1} \xi_2^a = i_{\xi_1} D \xi_2^a - i_{\xi_2} D \xi_1^a. \quad (\text{B1})$$

Using this, together with

$$(i_{\xi_1} R^a{}_b) \xi_2^b - (i_{\xi_2} R^a{}_b) \xi_1^b = i_{\xi_1} i_{\xi_2} D T^a + (i_{\xi_2} i_{\xi_1} R^a{}_b) e^b, \quad (\text{B2})$$

one finds, after some algebra, that

$$\begin{aligned} [\delta_{\text{LT}}(\xi_1), \delta_{\text{LT}}(\xi_2)] e^a &= i_{\xi_1} \mathcal{L}_{\xi_2} T^a - \mathcal{L}_{\xi_2} i_{\xi_1} T^a - (\mathcal{L}_{\xi_1} \xi_2^b) i_b T^a \\ &\quad - D(i_{\xi_2} i_{\xi_1} T^a) + (i_{\xi_2} i_{\xi_1} R^a{}_b) e^b - (i_{\xi_2} i_{\xi_1} T^b) i_b T^a. \end{aligned} \quad (\text{B3})$$

Moreover, the relation $[\mathcal{L}_X, i_Y] \alpha = i_{[X, Y]} \alpha$, where $\alpha \in \Omega^p(\mathcal{M})$, leads to

$$[\delta_{\text{LT}}(\xi_1), \delta_{\text{LT}}(\xi_2)] e^a = D(i_{\xi_1} i_{\xi_2} T^a) + (i_{\xi_1} i_{\xi_2} T^b) i_b T^a - (i_{\xi_1} i_{\xi_2} R^a{}_b) e^b, \quad (\text{B4})$$

which is expression given in the last commutator of Eq. (62).

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